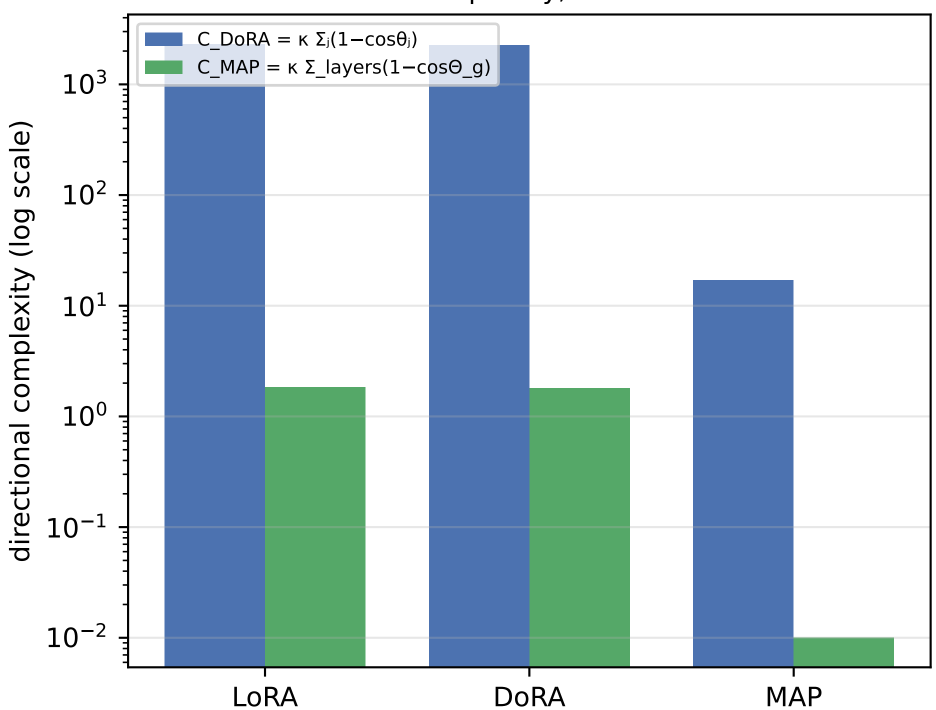
Directional Updates: LoRA vs DoRA vs MAP on RoBERTa-base / QQP

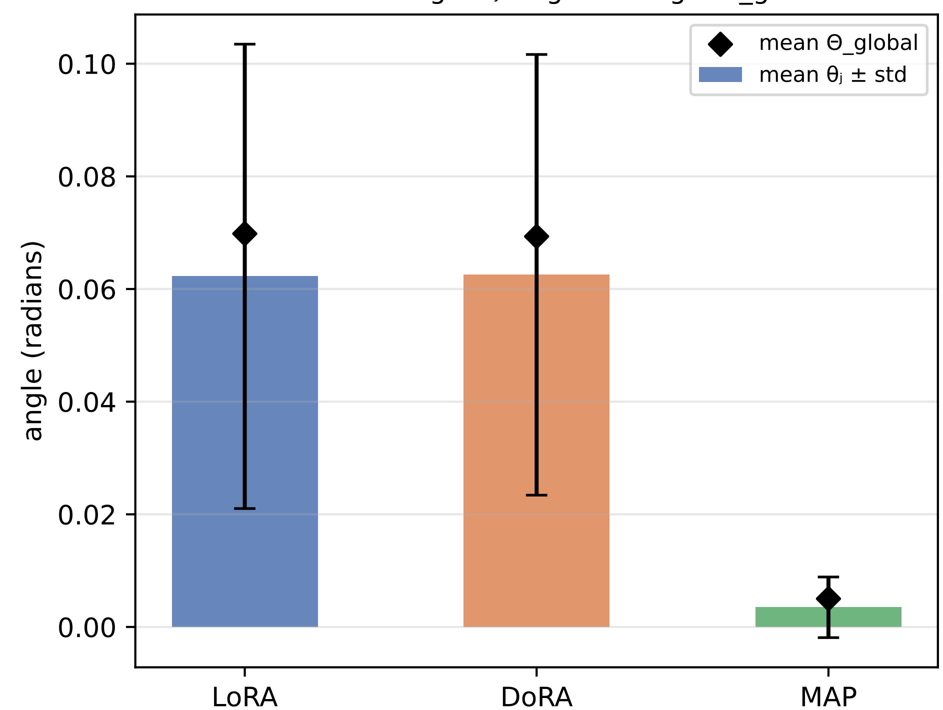
Fine-tuning metrics and aggregate directional complexity

Method	Eval acc	Trainable params	C_DoRA (Σ local)	C_MAP (global)	mean θ_j (rad)	mean Θ_g (rad)	mean s/d
LoRA	89.97%	1,919,234	2307.85	1.837	0.0622	0.0698	0.991
DoRA	90.05%	2,002,178	2252.28	1.801	0.0625	0.0693	0.997
MAP	86.88%	1,919,378	17.02	0.010	0.0035	0.0050	0.046

Local (sum) vs global (single rotation) complexity, κ=10

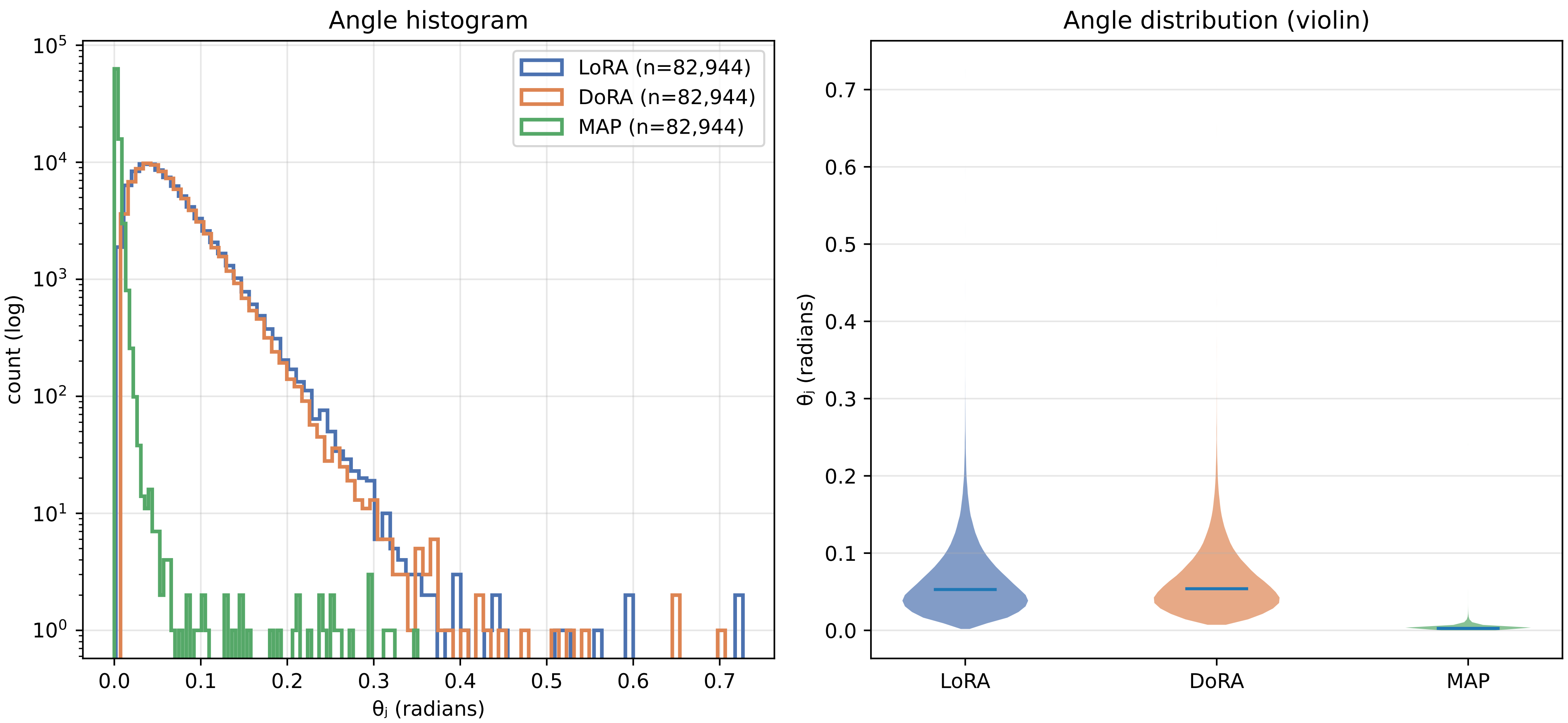


Per-column angle θⱼ vs global angle Θ_global

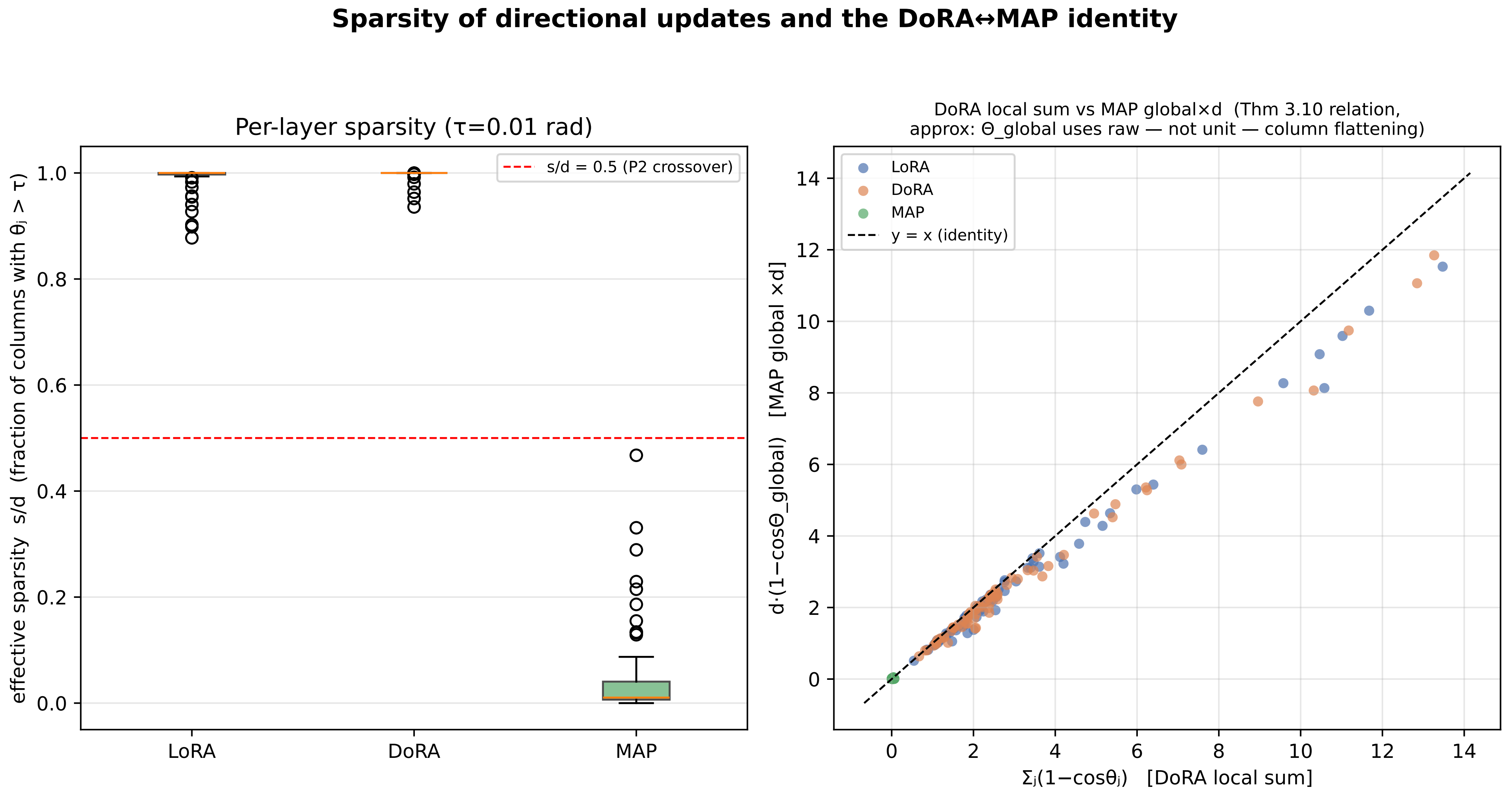


Θ_global is the single rotation of the whole flattened matrix (MAP view); θ_j are the per-column rotations (DoRA view).
Exact identity (Thm 3.10): $1-\cos\Theta$_global = mean_j$(1-\cos\theta_j)$ → DoRA sums what MAP averages.

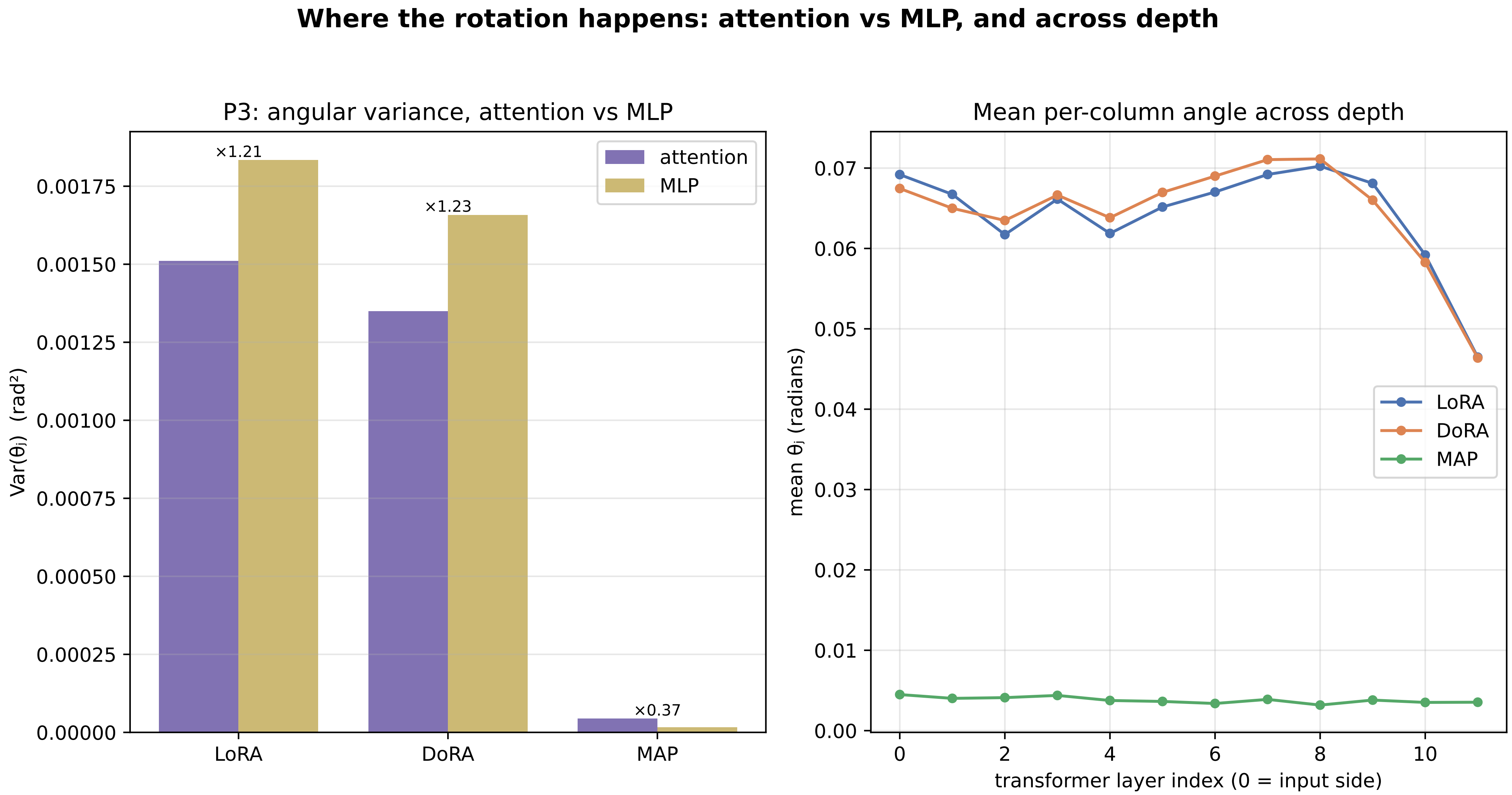
Distribution of per-column directional change θ_j (pooled over all target layers)



A long right tail = a few columns rotate a lot (sparse/localized update). A concentrated body near 0 = most columns barely move.



Sparsity of directional updates and the DoRA↔MAP identity
Per-layer sparsity (τ=0.01 rad)
s/d = 0.5 (P2 crossover)
effective sparsity s/d (fraction of columns with θj > τ)
1.0
0.8
0.6
0.4
0.2
0.0
LoRA
DoRA
MAP
DoRA local sum vs MAP global×d (Thm 3.10 relation,
approx: Θ_global uses raw — not unit — column flattening)
LoRA
DoRA
MAP
y = x (identity)
d·(1−cosΘ_global) [MAP global ×d]
Σj(1−cosθj) [DoRA local sum]
0
2
4
6
8
10
12
14



Where the rotation happens: attention vs MLP, and across depth
P3: angular variance, attention vs MLP
attention
MLP
×1.21
×1.23
×0.37
Var(θj) (rad²)
0.00000
0.00025
0.00050
0.00075
0.00100
0.00125
0.00150
0.00175
LoRA
DoRA
MAP
Mean per-column angle across depth
LoRA
DoRA
MAP
mean θj (radians)
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0
2
4
6
8
10
transformer layer index (0 = input side)